Bayesian Disease Mapping with R-INLA

How much should we trust a number?

Eric M. Delmelle · Lehigh University

Today

Some numbers rest on a lot of data, some on very little. Today we learn the Bayesian way of taking that into account, first with restaurants and a well, then with lung cancer in Pennsylvania. Reading: **Moraga Ch. 3, 4 and §5.1–5.3**.

> By the end of today you can
>
> 1. explain *prior*, *data* and *posterior* with an everyday example;
> 2. explain why the SIR of a small county is unreliable;
> 3. say what *shrinkage* does, and to which counties;
> 4. compute a 95% credible interval by hand, and read one off a posterior curve;
> 5. read a smoothed map, credible intervals and an exceedance map made with R-INLA.

```
library(sf)
library(dplyr)
library(tidyr)
library(ggplot2)
library(patchwork)
library(spdep)
library(SpatialEpi)   # pennLC
library(INLA)
```

> Warning
>
> **Install INLA before class.** It is not on CRAN:
>
> ```
> install.packages("INLA",
>   repos = c(getOption("repos"), INLA = "https://inla.r-inla-download.org/R/stable"))
> ```

The Bayesian idea in one example

Two restaurants

	Number of reviews	Average rating
Restaurant A	2	5.0
Restaurant B	800	4.7

Which one would you pick? Most people pick B, even though A has the higher average: two five-star reviews could come from the owner's friends. Without noticing, you did three things.

1. You started from what you know about restaurants in general: a typical one gets about 4.0 stars. That is the **prior**.
2. You looked at each restaurant's own reviews. That is the **data**.
3. You combined the two, trusting the reviews more when there are many. That is the **posterior**, your best guess.

The best guess is a weighted average

Say our prior is worth 10 reviews of 4.0 stars. The best guess for a restaurant is

$$\text{best guess} \;=\; w \times \text{its average} \;+\; (1-w) \times 4.0, \qquad w = \frac{\text{its reviews}}{\text{its reviews} + 10}.$$

w is how much we **trust its own reviews**, between 0 and 1. In R:

```r
best_guess <- function(reviews, average, prior_mean = 4.0, prior_worth = 10) {
  w <- reviews / (reviews + prior_worth)        # trust in its own reviews
  w * average + (1 - w) * prior_mean            # weighted average
}

best_guess(reviews = 2,   average = 5.0)        # Restaurant A
```

```
[1] 4.166667
```

```r
best_guess(reviews = 800, average = 4.7)        # Restaurant B
```

```
[1] 4.691358
```

Restaurant A's 5.0 is pulled back to 4.17: two reviews are not enough to convince us. B barely moves, to 4.69. Pulling an estimate toward the average is called **shrinkage**.

More reviews, less shrinkage

What if A kept getting five-star reviews?

```r
tibble(reviews = c(1, 2, 5, 10, 50, 100, 800)) |>
  mutate(w          = round(reviews / (reviews + 10), 2),   # trust in its reviews
         best_guess = round(best_guess(reviews, 5.0), 2))
```

```
# A tibble: 7 x 3
  reviews     w best_guess
    <dbl> <dbl>      <dbl>
1       1  0.09       4.09
2       2  0.17       4.17
3       5  0.33       4.33
4      10  0.5        4.5
5      50  0.83       4.83
6     100  0.91       4.91
7     800  0.99       4.99
```

```r
tibble(reviews = 1:1000) |>
  mutate(guess = best_guess(reviews, 5.0)) |>
  ggplot(aes(reviews, guess)) +
  geom_hline(yintercept = c(4, 5), linetype = 2, colour = "grey50") +
  annotate("text", x = 1, y = c(3.95, 5.05), hjust = 0, size = 3.5,
           label = c("prior: a typical restaurant (4.0)", "its own reviews (5.0)")) +
  geom_line(linewidth = 1.2, colour = "darkgreen") +
  geom_point(data = tibble(reviews = 2, guess = best_guess(2, 5.0)), size = 3) +
  annotate("text", x = 2.4, y = best_guess(2, 5.0), label = "Restaurant A",
           hjust = 0, vjust = 1.6, size = 3.5) +
  scale_x_log10() +
  labs(x = "Number of reviews (log scale)", y = "Best guess (stars)") +
  theme_classic(base_size = 12)
```

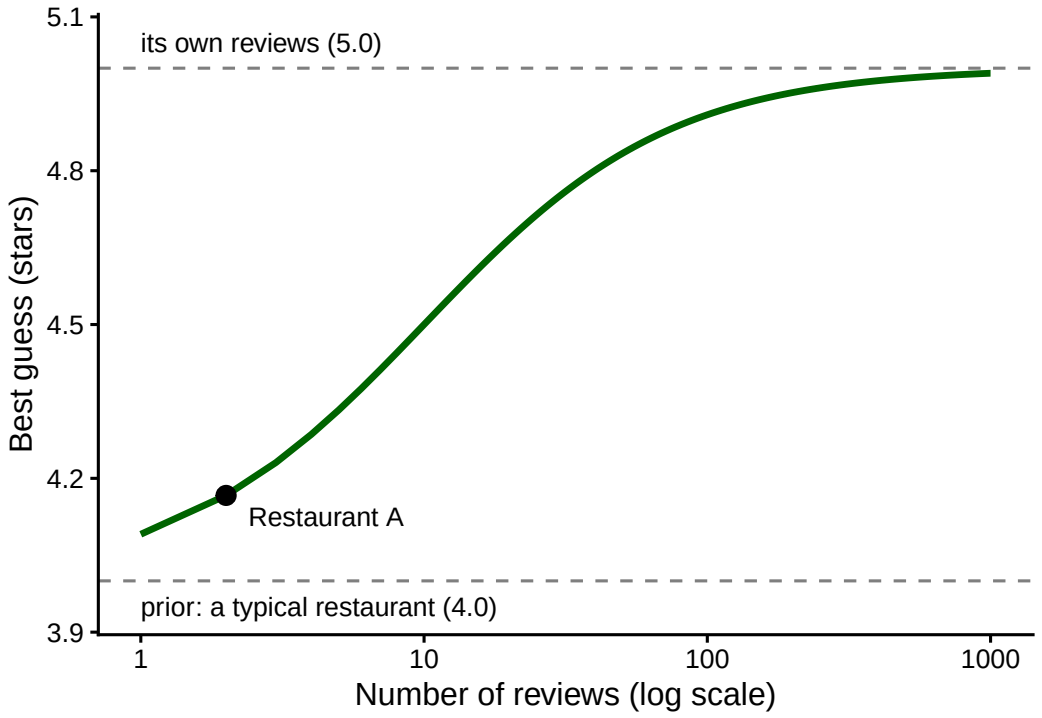


Figure 1: Best guess for a restaurant whose reviews all give 5.0 stars, as reviews accumulate.

With 2 reviews, the guess stays close to the prior. With 10 reviews, as many as the prior is worth, it sits exactly halfway. With hundreds, the prior hardly matters.

> Key idea
>
> The posterior is a compromise between the prior and the data. The more data, the more the data win.

Learning one test at a time: is the well water safe?

The story

A family tests its well for arsenic ten times. The EPA limit is 10 µg/L. Their question: **how often is this water above the limit?** Call that share θ: $\theta = 0.2$ means unsafe one test in five.

```
epa   <- 10                                      # EPA limit, µg/L
conc  <- c(14, 7, 12, 18, 11, 15, 9, 13, 14, 14)  # 10 tests (illustrative)
above <- as.integer(conc > epa)                  # 1 = above, 0 = below
above
```

```
 [1] 1 0 1 1 1 1 0 1 1 1
```

We do not start from nothing. In nearby wells, about **half** of past tests were above the limit: 5 of 10. That is our **prior**, centred on $\theta = 0.5$. A Beta curve keeps the score: Beta(5, 5) means 5 tests above and 5 below, borrowed from the neighbours. Each new test above the limit adds 1 to the first number, each test below adds 1 to the second.

Watching the posterior learn

In each panel, red dotted is what we believed **before** the test, green is what we believe **after**.

```
plot_update <- function(a, b, x, n, title = "") {
  theta <- seq(0, 1, 0.001)
  plot(theta, dbeta(theta, a, b), type = "l", lty = 3, lwd = 3, col = "red",
       ylim = c(0, 4), bty = "l", xlab = expression(theta), ylab = "",
       main = title)                                              # before
  if (n > 0)                                                      # no test, no update
    lines(theta, dbeta(theta, a + x, b + n - x), lwd = 3, col = "darkgreen")  # after
}
```

```
par(mfrow = c(3, 4), mar = c(4, 2, 2, 1))

a <- 5; b <- 5                                        # prior: 5 of 10 above in nearby wells
plot_update(a, b, 0, 0, title = "Before testing: nearby wells")

for (i in 1:10) {
  plot_update(a, b, above[i], 1,
              title = paste0("#", i, ": ", conc[i], " ug/L",
                             ifelse(above[i] == 1, " (above)", " (below)")))
  a <- a + above[i]                                   # after ...
  b <- b + 1 - above[i]                               # ... becomes next before
}
```

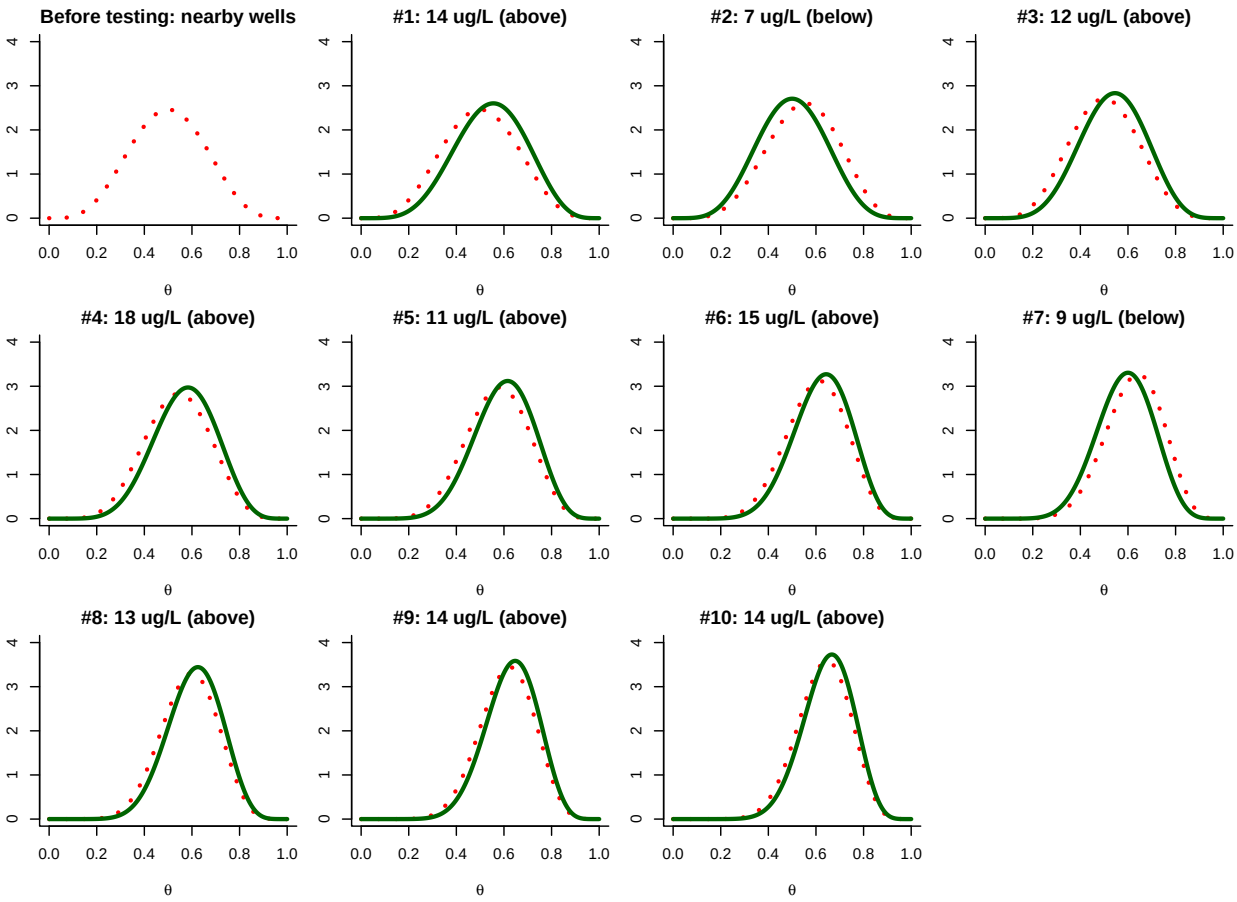


Figure 2: Each panel starts from the previous panel's green curve and adds one test.

Three things to notice:

1. A test above the limit pushes the curve right; a test below pushes it left (tests 2 and 7).
2. The curve gets **taller and narrower**: more tests, more certainty.
3. Each panel's green curve becomes the next panel's red curve. That is Bayesian learning.

> **Note**
>
> **Why does the y-axis stop at 4?** It shows a density, not a probability, so it can exceed 1; the area under each curve is 1. We fix it at 4 so the panels can be compared.

What do we know after ten tests?

```
a / (a + b)                                    # best guess for
```

```
[1] 0.65
```

```r
qbeta(c(0.025, 0.975), a, b)                    # 95% credible interval
```

```
[1] 0.4344984 0.8371141
```

```r
1 - pbeta(0.5, a, b)                            # P(unsafe more than half the time)
```

```
[1] 0.9164658
```

Our best guess is that the water is above the limit about 65% of the time, and with 95% probability between 43% and 84% (the **credible interval**). The probability that it is unsafe more than half the time is 0.92. Bayesian results answer the family's question directly, as a probability.

Where does 65% come from? The well's own tests say 8 of 10, 80%. The nearby wells say 50%. The prior is worth 10 tests and the well has 10 of its own, so the answer sits halfway. It is the restaurant formula again:

$$0.65 \;=\; w \times 0.80 \;+\; (1-w) \times 0.50, \qquad w = \frac{10}{10+10} = 0.5.$$

> Key idea
>
> We start from what we already know, here the nearby wells, and each new test updates it. More data, a narrower curve, more certainty.

Counties: why small counties need help

The data

We rebuild the Pennsylvania lung cancer counts from the SIR lecture: O observed cases, E expected cases, and SIR $= O/E$.

```r
data(pennLC)
d <- pennLC$data

ref <- d |>
  group_by(race, gender, age) |>
  summarise(r = sum(cases) / sum(population), .groups = "drop")   # reference rates

county <- d |>
  left_join(ref, by = c("race", "gender", "age")) |>
  group_by(county) |>
  summarise(O = sum(cases), E = sum(population * r), .groups = "drop") |>
  mutate(county = as.character(county), SIR = O / E)
```

```
pa <- st_as_sf(pennLC$spatial.polygon) |>
  st_set_crs(4326) |>
  mutate(county = names(pennLC$spatial.polygon)) |>
  st_transform(5070) |>                                          # Albers
  left_join(county, by = "county")

sir_breaks <- c(0, 0.67, 0.83, 0.95, 1.05, 1.2, 1.5, Inf)
sir_fill   <- scale_fill_brewer(palette = "RdBu", direction = -1, drop = FALSE, name = "Risk")
```

The SIR map

```
ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill +
  theme_void(base_size = 12)
```

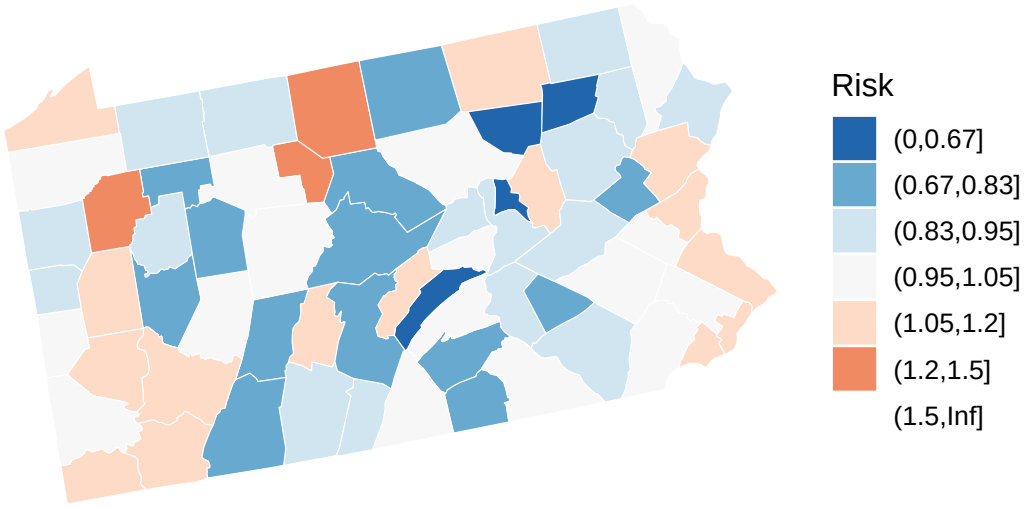


Figure 3: Lung cancer SIRs, Pennsylvania counties.

Which counties have the most extreme SIRs?

```
extremes <- county |>
  arrange(SIR) |>
  slice(c(1:3, 65:67)) |>                              # 3 lowest, 3 highest
```

```
  mutate(E = round(E, 1), SIR = round(SIR, 2))
extremes
```

```
# A tibble: 6 x 4
  county       O     E   SIR
  <chr>    <int> <dbl> <dbl>
1 juniata      6  18.7  0.32
2 sullivan     3   7.4  0.4
3 montour      8  16.1  0.5
4 cameron      8   5.9  1.35
5 venango     70  51.1  1.37
6 potter      22  16    1.37
```

Of these six counties, 5 expect fewer than 20 cases; the median county expects 75. A few cases more or less move their SIR a lot. Small counties are like restaurants with two reviews.

One more case

How fragile is a small county's SIR? Here is what **a single extra case** would do, in the three smallest and the three largest counties:

```
one_more <- county |>
  arrange(E) |>
  slice(c(1:3, 65:67)) |>                          # 3 smallest, 3 largest
  mutate(SIR_one_more = (O + 1) / E)               # with one more case

one_more |>
  mutate(E = round(E, 1), SIR = round(SIR, 2), SIR_one_more = round(SIR_one_more, 2))
```

```
# A tibble: 6 x 5
  county           O      E   SIR SIR_one_more
  <chr>        <int>  <dbl> <dbl>        <dbl>
1 forest           4    5.4  0.74         0.93
2 cameron          8    5.9  1.35         1.51
3 sullivan         3    7.4  0.4          0.54
4 montgomery     599  609.   0.98         0.99
5 allegheny     1275 1182.   1.08         1.08
6 philadelphia  1415 1219.   1.16         1.16
```

One more case would move Forest's SIR from 0.74 to 0.93, and Cameron's from 1.35 to 1.51. Forest and Cameron would even change colour on the map. Philadelphia's SIR would not move at two decimals. On the map, the colour of a small county can hinge on a single person.

The funnel plot

How far can an SIR stray by chance alone? If a county's true risk were exactly the state's, its SIR would still fall, about 95% of the time, within $1 \pm 1.96/\sqrt{E}$. Plotted against E, that range is a funnel.

```
funnel <- tibble(E = exp(seq(log(4), log(1500), length.out = 200))) |>
  mutate(lo = 1 - 1.96 / sqrt(E),                  # chance range of the SIR
         hi = 1 + 1.96 / sqrt(E))

labelled <- filter(county, county %in% c("sullivan", "potter", "juniata", "philadelphia"))

ggplot() +
  geom_ribbon(data = funnel, aes(E, ymin = lo, ymax = hi), fill = "grey85") +
  geom_hline(yintercept = 1, colour = "grey50") +
  geom_point(data = county, aes(E, SIR)) +
  geom_text(data = labelled, aes(E, SIR, label = tools::toTitleCase(county)),
            nudge_y = 0.07, size = 3.5) +
  scale_x_log10() +
  labs(x = "Expected cases (log scale)", y = "SIR") +
  theme_classic(base_size = 12)
```

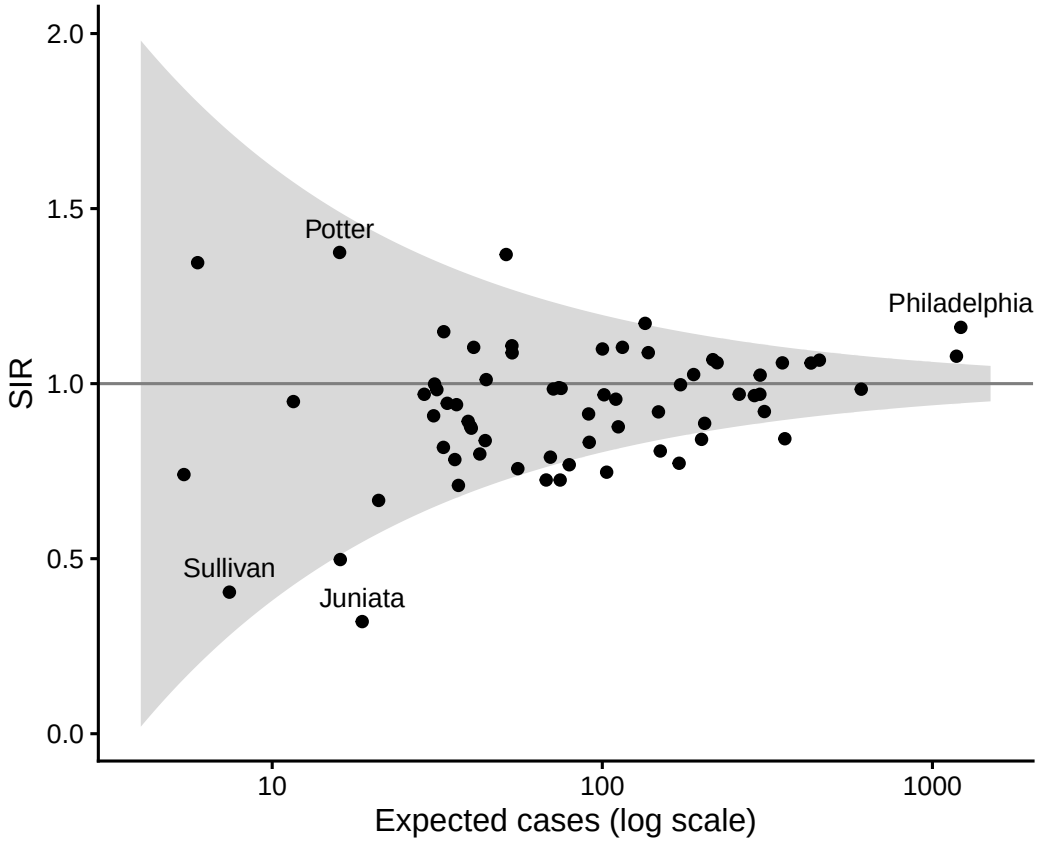


Figure 4: SIRs against expected cases. Grey: where about 95% of SIRs fall by chance alone.

The funnel is wide on the left. Potter's 1.37 and Sullivan's 0.40 sit inside it: they could be chance. Philadelphia's 1.16 is far outside: with more than 1,200 expected cases, a 16% excess is not chance.

> **!** The problem in one sentence
>
> A raw SIR mixes real differences in risk with chance, and chance is largest in small counties.

Shrinkage: the restaurant formula for counties

Same formula, new names

Restaurants	Counties
number of reviews	expected cases E
its average rating	its SIR
a typical restaurant, 4.0	the state average, 1
prior worth 10 reviews	state average worth k expected cases
trust in its own reviews, w	trust in its own SIR, w

So each county gets

$$\text{estimate} \;=\; w \times \text{SIR} \;+\; (1-w) \times 1, \qquad w = \frac{E}{E+k}.$$

A big county (large E) keeps its SIR. A small county borrows from the state.

Who decides k? The data

For the restaurants we picked 10 ourselves. For counties, R estimates k from the 67 SIRs:

- if counties differ **much more** than chance alone would produce, real differences are large, the state average is worth little, and k is small;
- if counties differ **about as much** as chance would produce, the state average is worth a lot, and k is large.

Letting the data choose the prior this way is called **empirical Bayes** (Clayton and Kaldor 1987; Marshall 1991). `EBest()` in **spdep** does it in one line.

```
eb <- EBest(pa$O, pa$E)                          # cases, expected cases
pa$RR_eb <- eb$estmm                             # shrunk estimate

k <- with(attr(eb, "parameters"), b / a)         # state worth k expected cases
k
```

```
[1] 107.908
```

```
pa$w <- pa$E / (pa$E + k)                                  # trust in own SIR
```

The state average is worth about 108 expected cases. Here are five counties, from small to large:

```
pa |>
  st_drop_geometry() |>
  filter(county %in% c("sullivan", "potter", "venango", "lehigh", "philadelphia")) |>
  arrange(E) |>
  transmute(county, E = round(E), SIR = round(SIR, 2),
            w = round(w, 2), estimate = round(RR_eb, 2))
```

```
        county    E  SIR    w estimate
1     sullivan    7 0.40 0.06     0.96
2       potter   16 1.37 0.13     1.05
3      venango   51 1.37 0.32     1.12
4       lehigh  260 0.97 0.71     0.98
5 philadelphia 1219 1.16 0.92     1.15
```

Sullivan expects 7 cases against the state's 108: we trust its SIR only 6%, and its estimate is almost the state's. Philadelphia, with more than 1,200 expected cases, keeps 92% of its SIR. Potter and Venango have the same SIR, 1.37, like two restaurants with the same average, but Venango rests on three times more cases, so it stays higher: 1.12 against 1.05.

Where do we borrow?

```
ggplot(pa) +
  geom_sf(aes(fill = w), colour = "white", linewidth = 0.2) +
  scale_fill_distiller(palette = "Purples", direction = 1, limits = c(0, 1),
                       name = "Trust w") +
  theme_void(base_size = 12)
```

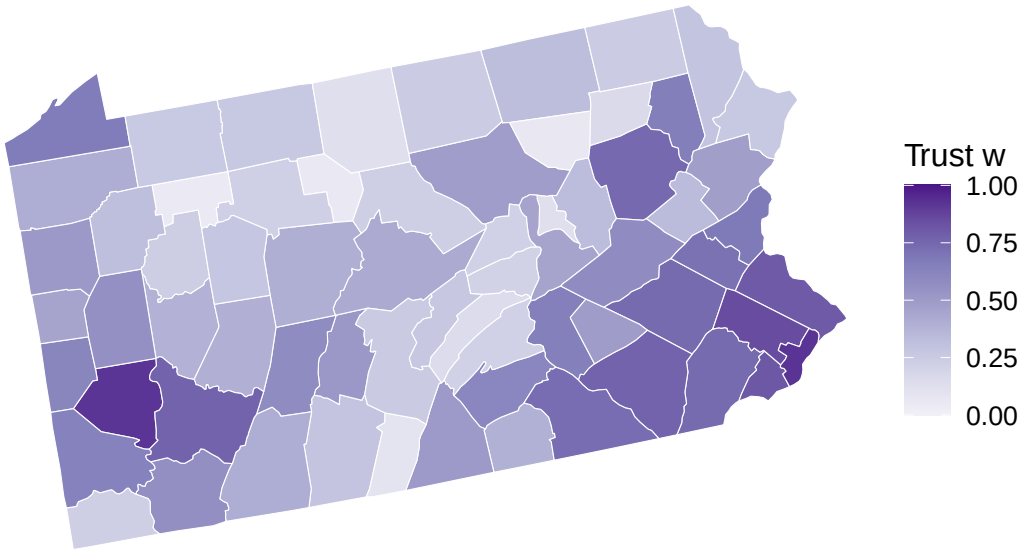


Figure 5: Trust in each county's own SIR. Dark: we keep the county's SIR. Light: we borrow from the state.

The map of w is a map of population: the rural north and centre borrow the most, the Philadelphia and Pittsburgh areas the least.

> **Key idea**
>
> Small counties borrow from the state; big counties keep their own SIR. The data decide how much to borrow.

From one number to a whole curve

Empirical Bayes gives one number per county, but behind it is a whole curve, as for the well. The state average, worth $k \approx 108$ expected cases, is the **prior**: a curve centred on 1 (red, dotted). The county's own O cases for E expected are the **data** (blue). The **posterior** combines the two (green). Its centre, the **best guess**, is the empirical Bayes estimate, $(k+O)/(k+E)$. The middle 95% of its area, shaded, is the **95% credible interval**: the county's true risk lies in it with 95% probability.

```
three <- county |>
  filter(county %in% c("sullivan", "potter", "philadelphia")) |>
  mutate(w          = E / (E + k),                    # trust in own SIR
         best_guess = (k + O) / (k + E),              # = empirical Bayes
         CI_low     = qgamma(0.025, k + O, k + E),    # 95% credible interval
```

```
         CI_high    = qgamma(0.975, k + O, k + E)) |>
  arrange(E)

three |>
  mutate(across(c(E, SIR, w, best_guess, CI_low, CI_high), \(x) round(x, 2)))
```

```
# A tibble: 3 x 8
  county           O       E   SIR     w best_guess CI_low CI_high
  <chr>        <int>   <dbl> <dbl> <dbl>      <dbl>  <dbl>   <dbl>
1 sullivan         3    7.42  0.4   0.06       0.96   0.79    1.15
2 potter          22   16     1.37  0.13       1.05   0.88    1.24
3 philadelphia  1415 1219.    1.16  0.92       1.15   1.09    1.21
```

```
curves <- function(name) {
  x <- filter(three, county == name)
  tibble(theta     = seq(0.01, 2.5, by = 0.002),
         Prior     = dgamma(theta, k, k),                # state, worth k cases
         Data      = dgamma(theta, x$O + 1, x$E),        # own cases only
         Posterior = dgamma(theta, k + x$O, k + x$E),    # both together
         county    = tools::toTitleCase(name))
}

cur <- bind_rows(lapply(three$county, curves)) |>
  pivot_longer(c(Prior, Data, Posterior), names_to = "curve", values_to = "density") |>
  mutate(curve  = factor(curve, c("Prior", "Data", "Posterior")),
         county = factor(county, tools::toTitleCase(three$county)))    # small to large
lines <- mutate(three, county = factor(tools::toTitleCase(county), levels(cur$county)))
shade <- cur |>
  filter(curve == "Posterior") |>
  left_join(select(lines, county, CI_low, CI_high), by = "county") |>
  filter(theta >= CI_low, theta <= CI_high)                          # middle 95%

ggplot(cur, aes(theta, density)) +
  geom_area(data = shade, fill = "darkgreen", alpha = 0.25) +
  geom_vline(data = lines, aes(xintercept = best_guess), colour = "darkgreen",
             linetype = 2) +
  geom_line(aes(colour = curve, linetype = curve), linewidth = 1) +
  facet_wrap(~ county, scales = "free_y") +
  scale_colour_manual(values = c(Prior = "red", Data = "blue", Posterior = "darkgreen")) +
  scale_linetype_manual(values = c(Prior = 3, Data = 1, Posterior = 1)) +
  labs(x = "Relative risk", y = NULL, colour = NULL, linetype = NULL) +
  theme_classic(base_size = 12) +
  theme(legend.position = "bottom")
```

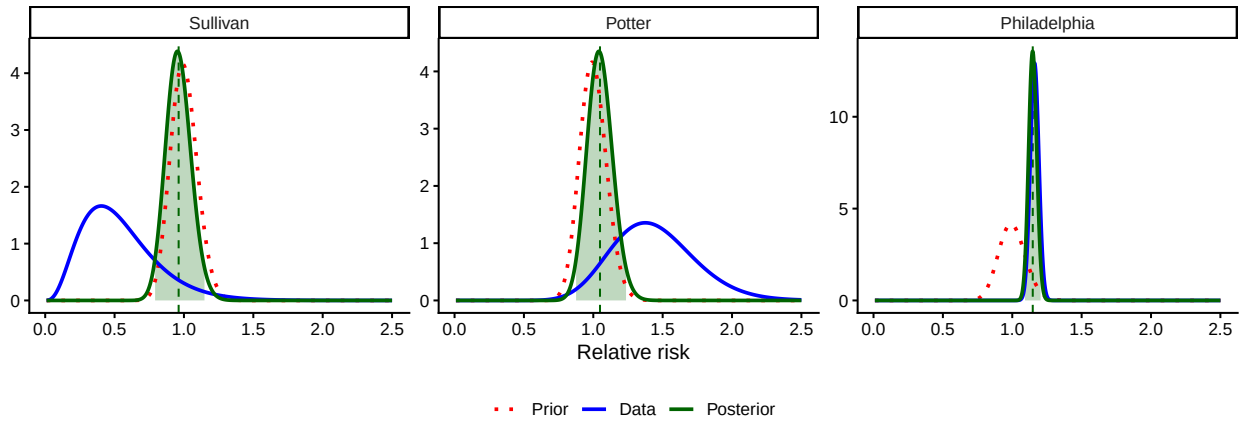


Figure 6: Prior (red), data (blue) and posterior (green) for a small, a middle and a large county. Shaded: the 95% credible interval. Dashed: the best guess.

Sullivan's green curve lies almost on the red prior: its 3 cases cannot outweigh 108 borrowed ones, and its interval, 0.79 to 1.15, is about as wide as the prior itself. Potter's SIR of 1.37 is pulled back to 1.05, and its interval, 0.88 to 1.24, contains 1: we cannot say that Potter's risk is above the state's. Philadelphia's curve lies on its own data, and its narrow interval, 1.09 to 1.21, lies entirely above 1.

The credible interval by hand

The posterior curve above is set by two numbers:

$$a = k + O \ \text{(borrowed + own cases)}, \qquad b = k + E \ \text{(borrowed + own expected cases)}.$$

With a calculator, four steps give the 95% credible interval:

Step	Formula
1. best guess	a/b
2. spread (standard deviation)	$\sqrt{a}\,/\,b$
3. low end	best guess $-\,1.96\times$ spread
4. high end	best guess $+\,1.96\times$ spread

This is the familiar "estimate $\pm$ 1.96 standard errors", applied to the posterior. R's `qgamma()` instead cuts exactly 2.5% of the area off each side of the curve. Both, for Sullivan:

```
su     <- filter(county, county == "sullivan")
a      <- k + su$O                                  # borrowed + own cases
b      <- k + su$E                                  # borrowed + own expected
best   <- a / b                                     # step 1
spread <- sqrt(a) / b                               # step 2
round(c(a = a, b = b, best_guess = best, spread = spread), 3)
```

```
          a           b best_guess      spread
    110.908     115.328       0.962       0.091
```

```r
round(best + c(-1.96, 1.96) * spread, 2)            # steps 3-4: by hand
```

```
[1] 0.78 1.14
```

```r
round(qgamma(c(0.025, 0.975), a, b), 2)             # exact: cut 2.5% each side
```

```
[1] 0.79 1.15
```

```r
ex   <- qgamma(c(0.025, 0.975), a, b)               # exact interval
hand <- best + c(-1.96, 1.96) * spread              # by hand
d    <- tibble(x = seq(best - 5 * spread, best + 5 * spread, length.out = 600),
               y = dgamma(x, a, b))
top  <- max(d$y)

ggplot(d, aes(x, y)) +
  geom_area(data = filter(d, x <= ex[1]), fill = "grey70") +                 # lower 2.5%
  geom_area(data = filter(d, x >= ex[2]), fill = "grey70") +                 # upper 2.5%
  geom_area(data = filter(d, x >= ex[1], x <= ex[2]), fill = "darkgreen", alpha = 0.25) +
  geom_line(colour = "darkgreen", linewidth = 1) +
  geom_vline(xintercept = best, colour = "darkgreen", linetype = 2) +         # best guess
  annotate("text", x = c(ex[1] - 0.6 * spread, best + 0.9 * spread, ex[2] + 0.6 * spread),
           y = top * c(0.12, 0.45, 0.12), label = c("2.5%", "95%", "2.5%")) +
  annotate("segment", x = hand[1], xend = hand[2], y = -0.12 * top,
           yend = -0.12 * top, linewidth = 1.2) +                            # by hand
  annotate("text", x = best, y = -0.22 * top, size = 3.5,
           label = "by hand: best guess +/- 1.96 x spread") +
  labs(x = "Relative risk, Sullivan County", y = NULL) +
  theme_classic(base_size = 12)
```

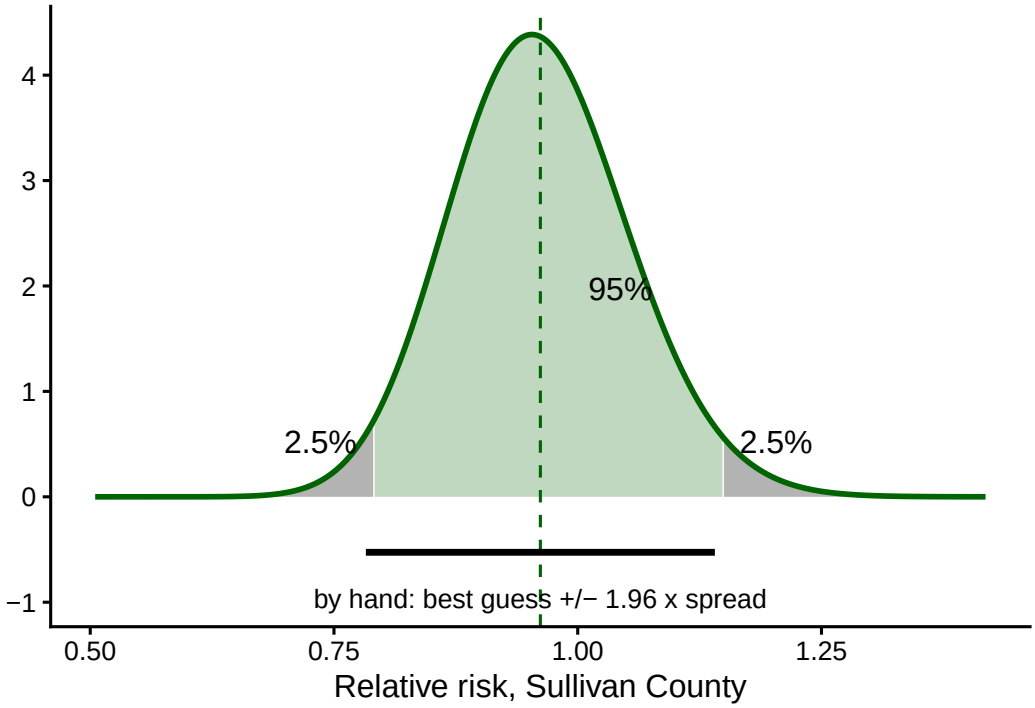


Figure 7: Sullivan's posterior. Grey: the 2.5% cut off on each side. Green: the middle 95%, the exact credible interval. Black bar: the interval by hand.

Here a is large, about 111, so the curve is almost symmetric and the two intervals agree to about 0.01. When a county borrows little, the curve is lopsided and the interval by hand is only an approximation; you will see this in Exercise 7.

Key idea

A credible interval is read off the posterior curve: cut 2.5% of the area on each side. By hand, best guess $\pm$ 1.96 $\times$ spread comes close.

Full Bayes with R-INLA

Empirical Bayes gives one number per county. A **full Bayesian** model does the same shrinkage but returns a whole posterior curve per county, like the green curves of the well. That tells us **how sure** we are. INLA is a fast way to fit such models (Moraga Ch. 3–4).

We use the **BYM2** model (Moraga §5.3). Instead of borrowing from the whole state, each county borrows mostly from its **neighbours**, the way the family borrowed from nearby wells. We use the queen neighbours from the Moran's I lecture.

How to read the `inla()` call below:

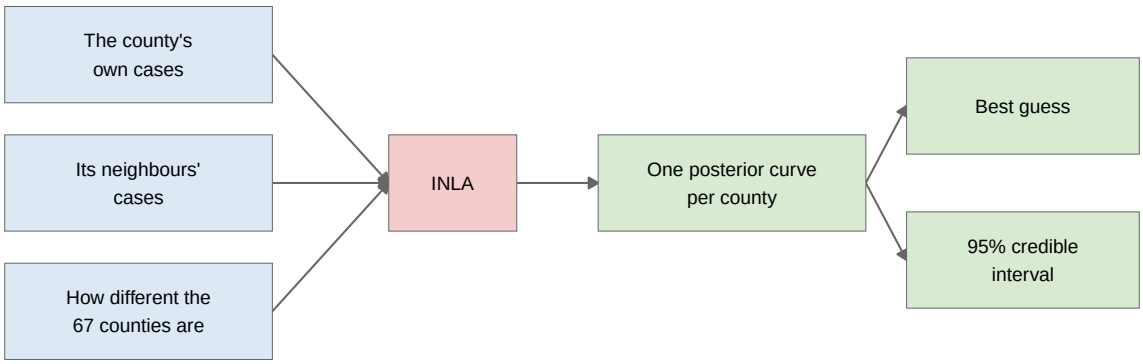


Figure 8: What goes into INLA, and what comes out.

Piece of the code	In words
`O ~ 1`	model the observed cases around one overall level of risk
`f(id, model = "bym2", graph = g)`	each county gets its own risk, pulled toward its neighbours (listed in `g`)
`family = "poisson", E = E`	cases are counts, compared with expected cases, so results read like an SIR
`control...`	only asks INLA to keep the results we need

```
dat <- st_drop_geometry(pa) |>
  mutate(id = row_number())                          # one index per county

nb <- poly2nb(pa, queen = TRUE)                      # queen neighbours
nb2INLA("pa.adj", nb)                                # write for INLA
g  <- inla.read.graph("pa.adj")

m2 <- inla(O ~ 1 + f(id, model = "bym2", graph = g),
           family = "poisson", E = E, data = dat,
           control.predictor = list(compute = TRUE),
           control.compute = list(return.marginals.predictor = TRUE))

dat$RR_bym2 <- m2$summary.fitted.values$mean         # posterior mean risk
dat$LL <- m2$summary.fitted.values$`0.025quant`      # 95% credible interval, low
dat$UL <- m2$summary.fitted.values$`0.975quant`      # 95% credible interval, high
```

Read the formula as: log-risk = an overall level + a bump for each county that borrows from its neighbours (Moraga §4.1).

Three maps

```
pa |>
  select(county) |>
  left_join(select(dat, county, SIR, RR_eb, RR_bym2), by = "county") |>
  pivot_longer(c(SIR, RR_eb, RR_bym2), names_to = "estimate", values_to = "risk") |>
  mutate(estimate = factor(estimate, c("SIR", "RR_eb", "RR_bym2"),
                           c("SIR (raw)", "Empirical Bayes", "BYM2 (INLA)"))) |>
  ggplot() +
  geom_sf(aes(fill = cut(risk, sir_breaks)), colour = "white", linewidth = 0.15) +
  facet_wrap(~ estimate) +
  sir_fill +
  theme_void(base_size = 12) +
  theme(legend.position = "bottom")
```

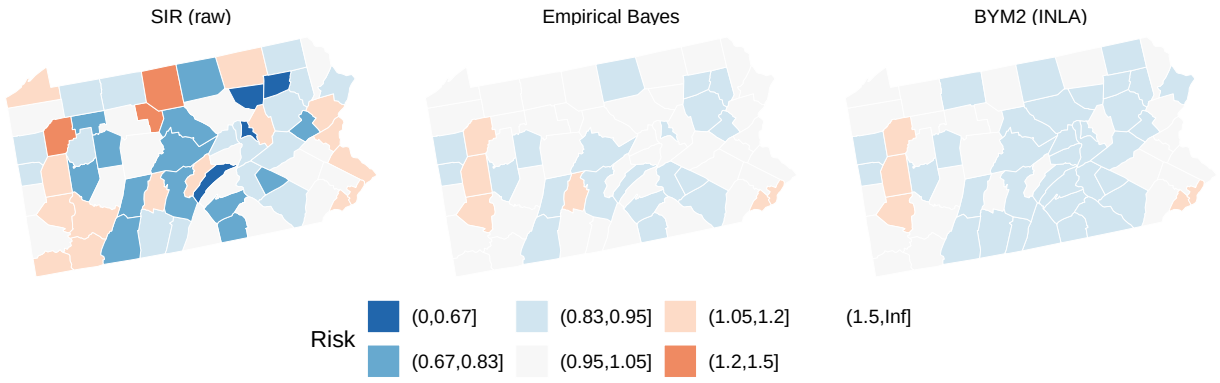


Figure 9: Raw SIR, empirical Bayes and BYM2. Same colours.

The extreme colours of the raw map, mostly small rural counties, fade. What remains is the broad pattern.

Two counties, same SIR

Greene and Philadelphia have almost the same SIR. Here are INLA's posterior curves for both, on the same horizontal axis. Green shaded: the 95% credible interval. Green dashed: the best guess. Black: the state average, 1.

```
post_curve <- function(name) {
  i  <- which(dat$county == name)
  cv <- inla.smarginal(m2$marginals.fitted.values[[i]])   # INLA's curve
  data.frame(county = paste0(tools::toTitleCase(name), ": SIR = ", sprintf("%.2f", dat$SIR[i])
```

```
                                ", E = ", round(dat$E[i])),
             x = cv$x, y = cv$y, RR = dat$RR_bym2[i], LL = dat$LL[i], UL = dat$UL[i])
}

cur2 <- bind_rows(post_curve("greene"), post_curve("philadelphia"))

ggplot(cur2, aes(x, y)) +
  geom_area(data = filter(cur2, x >= LL, x <= UL), fill = "darkgreen", alpha = 0.25) +
  geom_vline(aes(xintercept = RR), colour = "darkgreen", linetype = 2) +   # best guess
  geom_vline(xintercept = 1) +                                              # state average
  geom_line(colour = "darkgreen", linewidth = 1) +
  facet_wrap(~ county, scales = "free_y") +
  labs(x = "Relative risk", y = NULL) +
  theme_classic(base_size = 12)
```

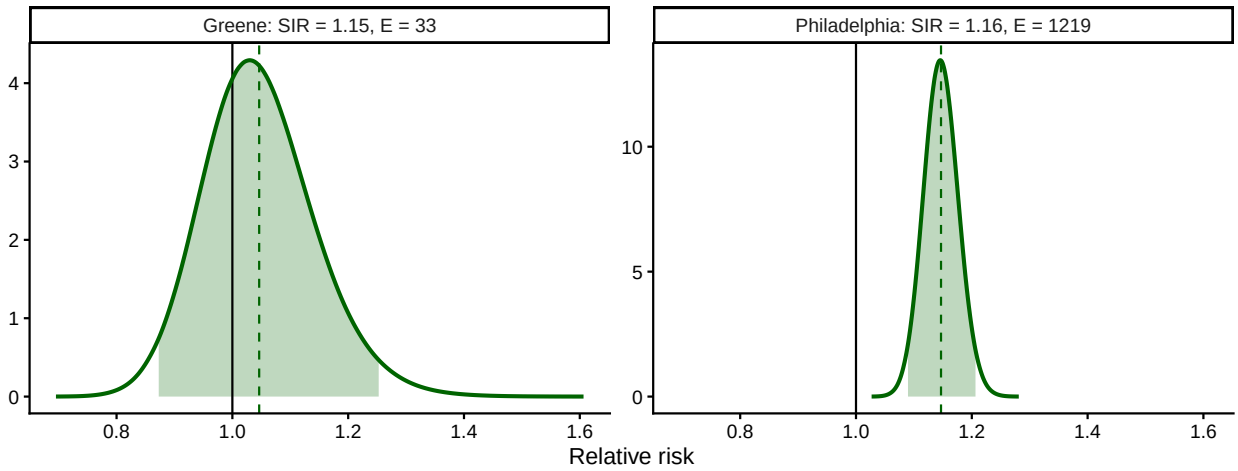


Figure 10: INLA's posterior curves for Greene and Philadelphia.

```
dat |>
  filter(county %in% c("greene", "philadelphia")) |>
  transmute(county, O, E = round(E), SIR = round(SIR, 2),
            RR_bym2 = round(RR_bym2, 2), LL = round(LL, 2), UL = round(UL, 2))
```

```
        county    O    E  SIR RR_bym2   LL   UL
1       greene   38   33 1.15    1.05 0.87 1.25
2 philadelphia 1415 1219 1.16    1.15 1.09 1.21
```

Greene's SIR rests on 33 expected cases: its curve is wide, and its interval, 0.87 to 1.25, contains 1. Philadelphia's rests on 1,219: its curve is narrow, and its interval, 1.09 to 1.21, lies entirely above 1. Same SIR, very different certainty.

All 67 counties at once

Each row is one county's posterior curve seen from above: the bar is its 95% credible interval, the dot its best guess, and the × its raw SIR. Small counties are at the top. Red: the whole interval is above 1. Blue: the whole interval is below 1. Grey: the interval contains 1.

```
ci_cols <- c("Entire interval above 1" = "#B2182B",
             "Interval contains 1"     = "grey55",
             "Entire interval below 1" = "#2166AC")

dat <- dat |>
  mutate(ci_class = factor(case_when(LL > 1 ~ names(ci_cols)[1],
                                     UL < 1 ~ names(ci_cols)[3],
                                     TRUE   ~ names(ci_cols)[2]), names(ci_cols)))

dat |>
  mutate(label = reorder(paste0(tc(county), " (E = ", round(E), ")"), -E)) |>
  ggplot(aes(y = label)) +
  geom_vline(xintercept = 1, linetype = 2) +                                  # state average
  geom_linerange(aes(xmin = LL, xmax = UL, colour = ci_class), linewidth = 1.2) +
  geom_point(aes(x = RR_bym2, colour = ci_class), size = 2) +
  geom_point(aes(x = SIR, shape = "Raw SIR"), size = 2) +
  scale_colour_manual(values = ci_cols, drop = FALSE, name = NULL) +
  scale_shape_manual(values = c("Raw SIR" = 4), name = NULL) +
  labs(x = "Relative risk", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", legend.box = "vertical")
```

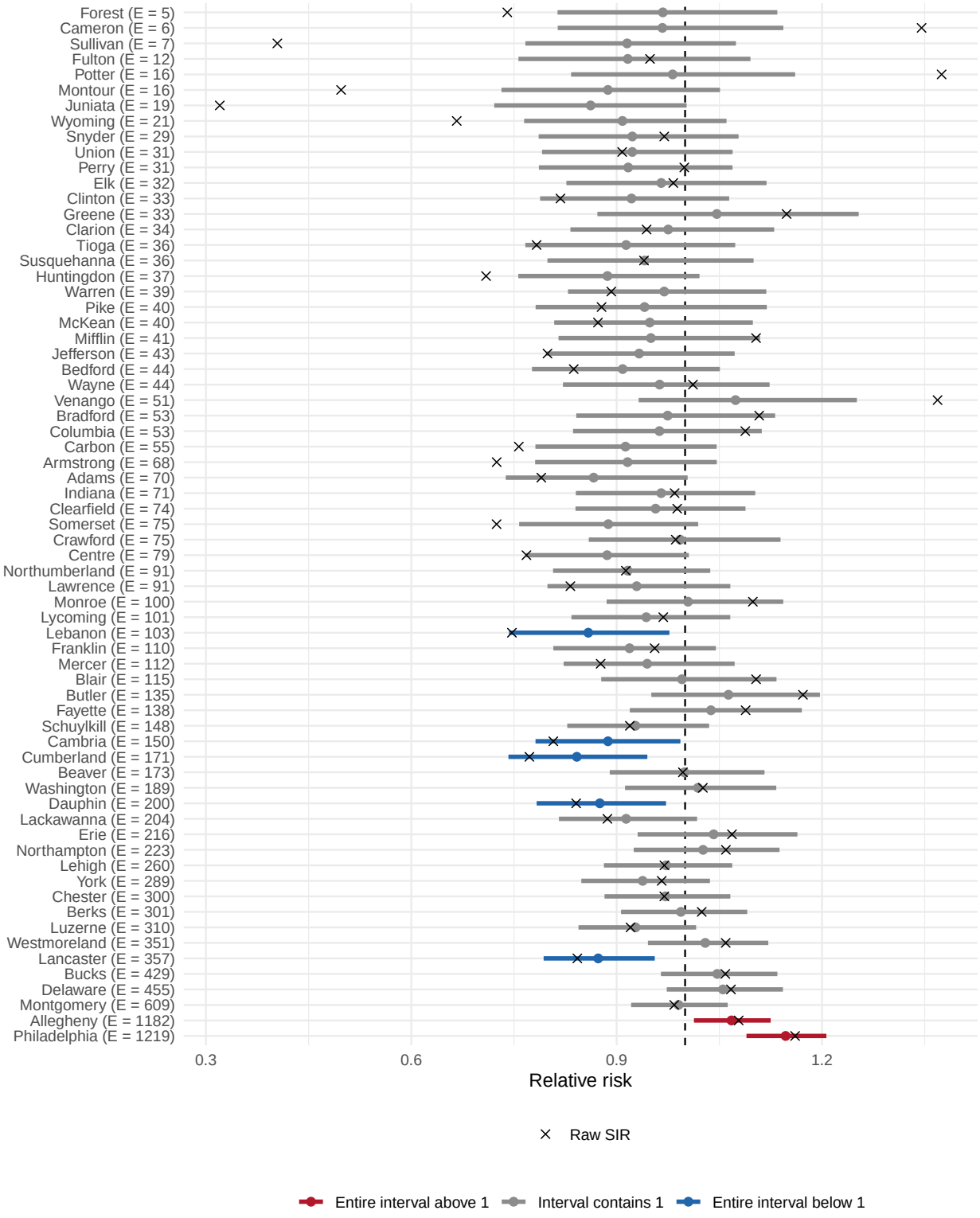


Figure 11: 95% credible intervals (BYM2) for all 67 counties, smallest at the top.

The same colours on the map:

```r
pa$ci_class <- dat$ci_class                              # same order as pa

ggplot(pa) +
  geom_sf(aes(fill = ci_class), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = ci_cols, drop = FALSE, name = NULL) +
  theme_void(base_size = 12)
```

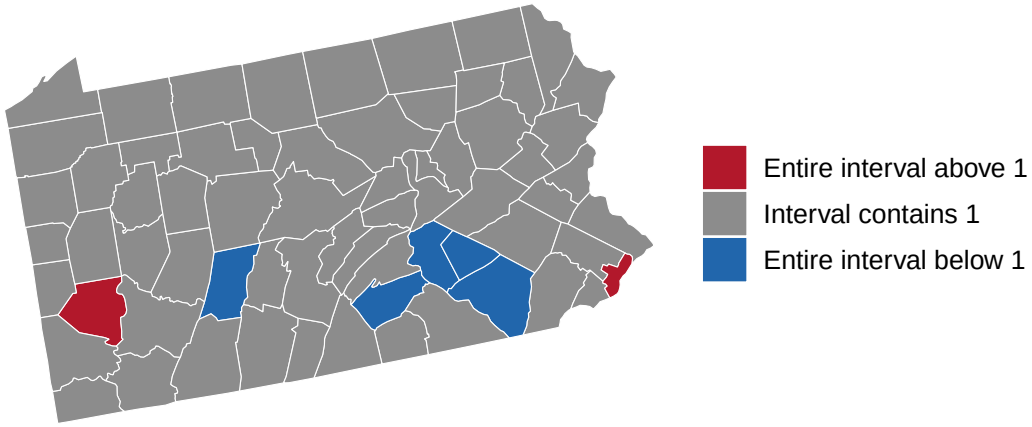


Figure 12: Is each county's 95% credible interval above or below the state average?

Small counties, at the top of the plot, have the longest bars, and their dots sit far from their ×: Juniata's SIR of 0.32 becomes 0.86. Only 2 counties have their whole interval above 1 (Allegheny, Philadelphia), and 5 have it below 1 (Cambria, Cumberland, Dauphin, Lancaster, Lebanon). For lung cancer, most counties are indistinguishable from the state average.

Borrowing from the neighbours

Empirical Bayes pulls every county toward the **state** average. BYM2 pulls it mostly toward its **neighbours**. Potter, outlined in black, and its neighbours:

```r
i    <- which(dat$county == "potter")
hood <- pa[c(i, nb[[i]]), ]                              # Potter, then its neighbours

p_map <- ggplot(hood) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white") +
```

```
  geom_sf(data = hood[1, ], fill = NA, colour = "black", linewidth = 1) +   # Potter
  geom_sf_text(aes(label = paste0(tc(county), "\n", sprintf("%.2f", SIR))),
               size = 3) +
  sir_fill +
  labs(title = "Raw SIRs") +
  theme_void() +
  theme(legend.position = "none")

rows <- c("Neighbours' SIRs", "State average", "Potter, raw SIR",
          "Potter, empirical Bayes", "Potter, BYM2 best guess")
pts <- tibble(what  = factor(c(rep(rows[1], nrow(hood) - 1), rows[2:5]), rev(rows)),
              value = c(hood$SIR[-1], 1, dat$SIR[i], dat$RR_eb[i], dat$RR_bym2[i]))
bar <- tibble(what = factor(rows[5], rev(rows)), lo = dat$LL[i], hi = dat$UL[i])

p_line <- ggplot(pts, aes(value, what, colour = what)) +
  geom_vline(xintercept = 1, linetype = 2, colour = "grey50") +
  geom_linerange(data = bar, aes(xmin = lo, xmax = hi, y = what), inherit.aes = FALSE,
                 colour = "darkgreen", linewidth = 1.2) +                    # credible interva
  geom_point(size = 3.5) +
  scale_colour_manual(values = c("grey40", "black", "blue", "purple", "darkgreen"),
                      breaks = rows, guide = "none") +
  labs(title = "Where does Potter land?", x = "Relative risk", y = NULL) +
  theme_minimal(base_size = 12)

p_map + p_line
```

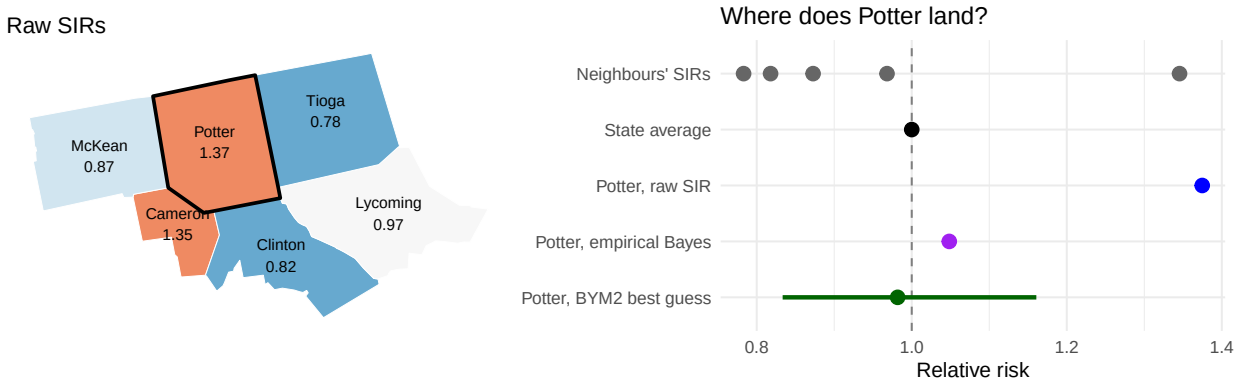


Figure 13: Left: raw SIRs of Potter and its neighbours. Right: where each method puts Potter; the green bar is its 95% credible interval.

Empirical Bayes pulls Potter from 1.37 to 1.05, toward the state average. BYM2 puts it at 0.98, closer to its neighbours: 4 of its 5 neighbours (Clinton 0.82, Lycoming 0.97, McKean 0.87, Tioga

0.78) have SIRs below 1. Cameron's SIR is high, 1.35, but it rests on only 6 expected cases, so it counts for little.

How sure are we? The exceedance map

For the well we asked: how likely is it that the water is unsafe more than half the time? For each county we ask: how likely is it that its risk is above the state average, $P(\theta > 1)$? This is the **exceedance probability**: the area of the posterior curve to the right of 1. A county whose whole 95% credible interval is above 1 has $P(\theta > 1) > 0.975$.

```
dat$p_exc <- sapply(m2$marginals.fitted.values,
                    function(m) 1 - inla.pmarginal(1, m))   # P( > 1)
```

```
exc_cols <- c("Probably lower (< 0.2)"  = "#2c7bb6",
              "Uncertain (0.2-0.8)"     = "grey90",
              "Probably higher (> 0.8)" = "#d7191c")

pa |>
  select(county) |>
  left_join(select(dat, county, p_exc), by = "county") |>
  mutate(exc = cut(p_exc, c(0, 0.2, 0.8, 1), labels = names(exc_cols),
                   include.lowest = TRUE)) |>
  ggplot() +
  geom_sf(aes(fill = exc), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = exc_cols, drop = FALSE, name = NULL) +
  theme_void(base_size = 12)
```

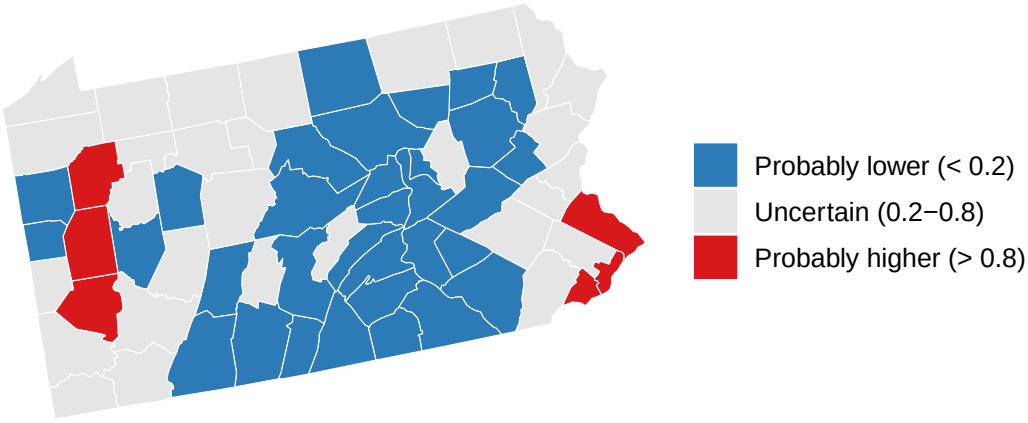


Figure 14: Probability that each county's lung cancer risk exceeds the state average (BYM2).

Counties probably above average: Allegheny, Bucks, Butler, Delaware, Philadelphia, Venango. Potter, despite its SIR of 1.37, is not among them ($P = 0.38$): 22 cases are not enough.

> Key idea
>
> INLA does the same shrinkage, borrows from the neighbours, and tells us how sure we are about each county.

In-class exercise

Work in pairs, 20 minutes.

1. **Restaurants.** Run `best_guess(2, 5.0, prior_worth = 100)`. What happens to A's best guess? And to B's, `best_guess(800, 4.7, prior_worth = 100)`? Why?
2. **Well water.** Suppose the family knew nothing about nearby wells and started from a flat prior, `a <- 1; b <- 1`. Rerun the updating. How does the final best guess change, and why?
3. **Funnel.** Cameron County has an SIR of 1.35 with only 6 expected cases. Is it inside the funnel? What is its empirical Bayes estimate, `pa$RR_eb[pa$county == "cameron"]`? Explain the difference.
4. **By hand.** Potter has $O = 22$ cases for $E = 16.0$ expected. With $k = 108$, compute with a calculator a, b, the best guess, the spread and the 95% credible interval by hand. Check with `qgamma(c(0.025, 0.975), 108 + 22, 108 + 16.0)`.

5. **Interpretation.** In two sentences for a county health officer, explain why Potter is not flagged even though its SIR is 1.37. Use its credible interval.

Key takeaways

1. The posterior is a compromise between the prior and the data; the more data, the more the data win.
2. A raw SIR mixes real differences and chance; chance is largest in small counties.
3. Shrinkage: small counties borrow from the state (or their neighbours), big counties keep their SIR.
4. A posterior is a curve: its centre is the best guess, its middle 95% the credible interval. By hand: best guess $\pm$ 1.96 $\times$ spread.
5. Report how sure we are, not only the estimate: a county is clearly above average only when its whole credible interval is above 1.

Words to know

Word	Plain meaning
Prior	what we believe before seeing the data
Likelihood	what the data say
Posterior	what we believe after seeing the data
Best guess	the centre (mean) of the posterior curve
Spread	the standard deviation of the posterior curve
Credible interval	a range that contains the true value with 95% probability
Shrinkage	pulling unreliable estimates toward an average
Empirical Bayes	shrinkage where the data decide how much to shrink
Exceedance probability	the probability that a risk is above a threshold, e.g. the state average

References

Clayton, D., & Kaldor, J. (1987). Empirical Bayes estimates of age-standardized relative risks for use in disease mapping. *Biometrics*, 43(3), 671–681.

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Riebler, A., Sørbye, S. H., Simpson, D., & Rue, H. (2016). An intuitive Bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4), 1145–1165.